

STUDENT	REGISTER NO.	ASSESSMENT
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AI RESEARCH PAPER ASSISTANT

Multi-Step Prompt Workflow for Literature Review, Claim Extraction & Critical Synthesis

01 | Concept

Purpose: AI Research Paper Assistant converts raw scientific PDF excerpts, research methodologies, experimental findings, and project abstracts into a structured literature review matrix and critical evaluation summary.

Problem: Students and undergraduate researchers are often assigned dozens of academic publications for their capstone projects. Synthesizing dense technical texts manually leads to cherry-picking biases, misinterpreting statistical claims, and struggling to structure a coherent Related Work chapter. A sequential AI prompt workflow structures this analysis into verifiable steps.

02 | Objective

The workflow executes sequential prompts directly in an AI chat interface to isolate foundational claims, benchmark empirical methodologies, cross-examine conflicting findings, generate a structured comparative literature table, and draft an academic critique with identified research gaps. Each stage depends strictly on the verified output of prior steps without code execution.

03 | Workflow Overview

INPUT	1	2	3	4	5	OUTPUT
Raw Papers & Abstracts	Claim Extraction	Method Audit	Cross Synthesis	Gap Spotting	Draft Section	Literature Review

04 | Workflow Steps

1. Core Contribution & Claim Extraction — Extract the core research question, proposed architecture/algorithm, and primary quantitative claims from paper abstracts. Output: Core Contribution Factsheet.

2. Methodology & Validation Audit — Audit experimental parameters, datasets used, benchmark baselines, and evaluation metrics (e.g., F1-score, latency, accuracy). Output: Methodology & Dataset Breakdown.

3. Comparative Cross-Paper Synthesis — Cross-compare the papers to identify agreements, algorithmic trade-offs, and conflicting empirical outcomes. Output: Comparative Synthesis Table.

4. Research Gap & Limitation Identification — Isolate unaddressed edge cases, scalability limitations, and computational bottlenecks noted across the evaluated studies. Output: Prioritized Research Gap Matrix.

5. Literature Review Drafting & Citation Synthesis — Draft a coherent, IEEE/ACM-styled 'Related Work' synthesis paragraph connecting the papers and positioning the student's project. Output: Formatted Related Work Section.

05 | Prompt Workflow (UI Execution Prompts)

Step 1 - Core Claim Extraction: Analyze the provided academic excerpts from the assigned papers. Extract the main research objective, proposed model or technique, and stated quantitative results. Present the findings as a bulleted summary. Do not infer conclusions not present in the text.

Step 2 - Methodological Audit: Using the Step 1 summary, inspect the methodology of each paper. Identify the training dataset, sample size, hardware or baseline models compared against, and primary performance metrics. Flag any missing benchmark details.

Step 3 - Comparative Cross-Paper Synthesis: Using the findings from Step 1 and Step 2, synthesize the approaches into a structured comparison. Highlight where Paper A outperforms Paper B, where their assumptions diverge, and what computational trade-offs exist between accuracy and speed.

Step 4 - Research Gap Analysis: Examine the Step 3 synthesis to identify technical blind spots. What edge cases, real-world constraints, or evaluation flaws did the authors leave unresolved? List 3 distinct research gaps that a capstone project could address.

Step 5 - Related Work Synthesis Drafting: Using the Step 3 comparative findings and Step 4 research gaps, draft a formal 2-paragraph 'Related Work' section in academic tone. Contextualize how the evaluated literature motivates our proposed capstone architecture.

06 | Prompt Chaining Architecture

Step 1 establishes the verified factual claims; Step 2 evaluates validation rigor; Step 3 combines both to map trade-offs; Step 4 isolates open limitations; and Step 5 unifies all insights into a publication-grade literature review section.

STAGE	INPUT / CONTEXT INGESTED	OUTPUT PRODUCED
1	Raw Paper Abstracts & Introductions	Extracted Contributions & Quantitative Claims
2	Step 1 Claims + Method Sections	Methodology, Baseline & Dataset Audit
3	Step 1 Claims + Step 2 Method Audit	Cross-Paper Comparative Trade-Off Matrix
4	Step 3 Comparative Matrix	Ranked Research Gaps & Unresolved Limitations
5	Step 3 Matrix + Step 4 Research Gaps	Academic-Ready 'Related Work' Draft Section

07 | Demonstration Input

Scenario: A final-year student researching 'Lightweight Deep Learning for Edge Healthcare Devices' reviews three recent papers comparing CNN pruning, 8-bit post-training quantization, and knowledge distillation on the MIT-BIH Arrhythmia ECG dataset.

08 | Expected Final Output

A cohesive literature analysis: an empirical trade-off table comparing memory footprint vs. F1-score across quantization and distillation, an identified research gap in real-time edge drift adaptation, and a formal 2-paragraph Related Work writeup ready for the final thesis.

09 | Workflow Result & Evaluation

The completed workflow transforms unstructured technical reading into a rigorous, connected literature review pipeline. It prevents hallucinated citations and ensures that thesis claims are supported by concrete comparative analysis.

10 | Sample Final Research Synthesis Plan (Step-by-Step Milestone)

PHASE	RESEARCH TARGET	PLANNED ACTIVITY	DELIVERABLE / OUTPUT
Phase 1	Paper Extraction	Extract claims and architectures from 3 edge AI papers	Structured Claim Factsheet
Phase 2	Method Audit	Compare MIT-BIH dataset splits and inference metrics	Methodology & Baseline Matrix
Phase 3	Comparative Matrix	Map parameter reduction vs. arrhythmia F1 drop	Cross-Study Trade-Off Table
Phase 4	Gap Isolation	Identify lack of dynamic integer quantization under battery drain	Ranked Research Gap Summary
Phase 5	Thesis Drafting	Draft Related Work chapter synthesizing trade-offs & gap	Polished 2-Paragraph Thesis Section

11 | Key Benefits

- Structured claim extraction avoids subjective reading and selective citation bias
- Systematic methodological auditing ensures direct comparison across identical datasets
- Clear separation of extraction, synthesis, and drafting minimizes AI hallucination
- Directly yields an empirical trade-off matrix suitable for project reports and defense presentations
- 100% zero-code workflow executed entirely within standard AI chat interfaces

12 | Conclusion

AI Research Paper Assistant illustrates how multi-step prompt chaining streamlines scholarly literature reviews. By decomposing a complex academic task into sequential stages—claim isolation, methodology benchmarking, comparative synthesis, gap discovery, and academic drafting—the workflow enables students to produce rigorous, audit-ready research syntheses without manual fragmentation.